Pablo Pena

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EDUCATION

University of Texas at Austin August 2024 – May 2028
B.S. in Mechanical Engineering Honors, Robotics Minor GPA: 4.0

SKILLS

CAD & Analysis: SolidWorks, Siemens NX, FEA, GD&T
Manufacturing: 3D Printing, Laser Cutting, SMT Soldering, Wire Harness Fabrication, CF Layups
Programming: Python, MATLAB, C#, Arduino, SQL, Java

PROFESSIONAL EXPERIENCE

BlackPearl Technology June 2026 - August 2026
Mechanical Engineering Intern

- Reduced component count by 67% through design consolidation, streamlining manufacturability and reducing assembly complexity
- Reduced product assembly times by 60% through fixture and process improvements
- Designed and documented 20+ components and assemblies for production and testing applications using SolidWorks

Landry's Inc. June 2024 - August 2024
Information Technology Intern

- Developed a Power BI dashboard consolidating 9 key IT performance metrics, improving visibility and streamlining reporting across departments
- Reviewed and corrected 180+ intranet entries to improve data accuracy and maintain up-to-date operational information across restaurant locations
- Developed a C# application leveraging APIs to automate data transfers between loyalty systems, eliminating manual CSV processing

PROJECTS

RC Plane — Texas Design Build Fly August 2024 - Present

- Designed and fabricated 4 fuselage prototypes using hybrid composite layups and vacuum bagging, optimizing stiffness-to-weight ratio and manufacturability
- Manufactured landing gear with carbon fiber/Kevlar with ±45° CF plies for increased torsional stiffness and 0° Kevlar plies for deflection resistance; validated layup quality through fit checks and assembly testing

Battle Bot January 2025 - May 2025

- Designed and fabricated 10 custom components using SolidWorks, 3D printing, and laser cutting to create a sub-1 lb combat robot
- Selected and integrated drive, actuation, and power subsystems, including DC motors, servo actuation, compression springs, and a 7.4 V LiPo battery pack, to improve durability and weapon performance

ACHIEVEMENTS & AWARDS

- 6th Place Overall — AIAA Design/Build/Fly International Competition April 2025
- Hispanic Scholarship Fund (HSF) Scholar December 2024